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Lab1: Your First C++ Program

Objective(s)

1. Learn how to use Visual Studio.
2. Learn about Input/output statements in C++ and how to use them in problem solving.
3. Learn about basic data types in C++ and how to use them in problem solving.

Tool(s)/Software

Visual Studio.

Description

- Learning how to use Visual Studio to write, save, compile and execute C++ programs.
- Learning how to find and fix the errors in Visual Studio.
- Writing C++ programs with reading/ writing statements.
- Writing C++ programs with different data types.
- Writing C++ programs with mathematical expressions.

Tasks/Assignments(s)

For each of the following problems: Fill the following IPO chart (along with pseudocode) and draw the flowchart that represents its logic. Then write a C++ program for that task.

Task#1 Problem:

Create a program that allows the user to enter two values. The program outputs the product of the two values.

Input	Processing	Output
	Processing items: *Algorithm:*	

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Task#2 Problem:

Create a program that allows the user to enter values for a salesperson's base salary, total sales, and commission rate. The program computes and outputs the salesperson's pay by adding the base salary to the product of the total sales and commission rate.

Input	Processing	Output
	Processing items: *Algorithm:*	

Task#3 Problem:

a) Create a program that allows the user to enter values for the width and length of a room's floor in feet. The program outputs the area of the floor in square feet.
b) Modify the program that computes floor area to compute and output the number of 6-inch square tiles needed to tile the floor. Then write a C++ program for this task.

Input	Processing	Output
	Processing items: *Algorithm:*	

Post-Lab Exercise:

Create a program that allows the user to enter a value for one edge of a cube. The program calculates the surface area of one side of the cube, the surface area of the cube, and its volume. The program outputs all the results.

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